

Demo Pack: Enterprise Semantic Graph for Product Development

Scenario: a Product Manager designs the 'FedEx view' Wire Tracking Center with an agent grounded in the Payments BU semantic graph - storyline, 10 questions with graph-grounded answers, effort plan and answer key

Document ID	DEMO-SG-WTC-1.0
Version / Status	1.0 / Demo-ready
Document owner	Demo team
Approver(s)	-
Effective / Last reviewed	2026-10-08
Classification	DEMO MATERIAL
Corpus	15 source documents (section 2.1) + the PM upload PM-WTC-MR-0.3
Agent	Chat agent with Anthropic product-management plugin skills + custom 'estate-discovery' skill + graph tools (wiki_read, graph_query)

1. Demo at a glance

The claim: product discovery fails less on ideas than on *enterprise context* - who owns what, what actually exists, what was tried before, and which rules apply. That context is scattered across architecture documents, API catalogs, policies, Jira and memos. No single document contains the answer; the answer lives in the **connections**. A semantic graph makes those connections traversable, so an agent can do in minutes what normally takes weeks of meetings.

The scenario: Sarah Whitfield (Senior PM) wants to add a FedEx-style Wire Tracking Center with predictive insights to the existing Crestline Business Online portal. She uploads her market research and concept (PM-WTC-MR-0.3) and asks the agent to help her write a BRD. The concept looks reasonable - and contains four serious problems that are invisible from any single document.

#	Headline reveal	What the PM assumed	What the graph shows
1	The tooltip that would tip off	Show exact hold reasons; Confirm/Reject on all held wires	Violates POL-FCC-014 and SAR confidentiality - and story CBO-4388 is about to ship raw risk text to production on 2026-10-22
2	'Settled' is not a field	'Completed' already means settled	'Completed' is shown at release (v1 collapses 4 states). Needs v2 migration with a hard 2027-03-31 deadline, rail-aware semantics and a backlog API change with no sponsor
3	The FedEx map has no live feed - and a new owner	Live gpi route via Raj Malhotra's team	gpi data is a 4-hourly batch with no API; quota blocks per-wire calls; ARB forbids direct exposure; ownership moved to GTSI on 2026-10-01
4	The insight that uses someone else's bank account	'Beneficiary typically moves funds within 4 hours'	Only derivable from another client's deposit activity in a Restricted, fraud-only feature table - a cross-client confidentiality and model-use violation

2. Demo setup

2.1 Source corpus (ingested into the graph)

Doc ID	File name	Contributes to the graph
CNB-ORG-PTT-2026-06	CNB-ORG-PTT-2026-06 PTT Organization System Ownership Engagement Directory	Teams, leaders, system ownership (as of June), forums, planning factors, PI calendar
CNB-MEMO-2026-09	CNB-MEMO-2026-09 Realignment of Cross-Border Network Integration	Ownership change PNE -> GTSI effective 2026-10-01 (temporal override)
CBO-ARCH-WC-4.1	CBO-ARCH-WC-4.1 Wire Center Current-State Architecture	Existing portal: integrations (PPH v1), status-label mapping, identifiers, notifications, reusable assets
PPH-SYS-OVW-9.2	PPH-SYS-OVW-9.2 PRISM Payments Hub System Overview and Lifecycle State Model	Lifecycle states, v1 collapse, rail-specific NETWORK_ACCEPTED semantics, HMS description sync
PPH-API-CAT-2026.3	PPH-API-CAT-2026.3 PRISM Payments Hub API and Event Catalog	v1/v2 endpoints, fields, v1 sunset, consumer migration status, topics, backlog

Doc ID	File name	Contributes to the graph
PNG-TDD-6.0	PNG-TDD-6.0 Payment Network Gateway Fedwire and gpi Technical Design	Fedwire acks/OMAD, gpi batch integration, quota, G001 behavior, outcome stats (owner shown pre-reorg)
PRSP-HMS-3.4	PRSP-HMS-3.4 Hold Management Service Design and Reason Taxonomy RESTRICTED	Hold codes, tiers, Restricted descriptions, disposition API, controls CTRL-PAY-012/018/031/040
ENS-INT-3.2	ENS-INT-3.2 Enterprise Notification Service Integration Guide	Event catalog, TPL-WIRE-REL-02 wording, subscription limits, onboarding SLA
TDIP-CAT-2026.2	TDIP-CAT-2026.2 Treasury Data and Insights Platform Data Catalog Extract	Datasets, classifications, beneficiary_behavior_profile, corridor stats, Insights API, gaps
POL-FCC-014	POL-FCC-014 v3.2 Customer Communication of Payment Status Holds and Exceptions Standard	Disclosure tiers, indistinguishability, status terminology, sharing, insight rules, copy library
DUS-07	DUS-07 v2.1 Data Use and Client Confidentiality Standard	Purpose limitation, cross-client rule, cohort thresholds, own-data rules
MRM-POL-02	MRM-POL-02 v7.0 Model Risk Management Policy Extract	Model vs EUA, Tier 2 for customer-facing estimates, use limitations, queue
JIRA-EXP-2026-10-05	Jira Export WT-discovery 2026-10-05	29 issues incl. CBO-4388 (in flight), CBO-4471, PPH-2207, FCT-1893, comments, CMP-2026-1189
ARB-PAY-REG-2026Q3	ARB-PAY-REG-2026Q3 Payments ARB ADR Register and Minutes	ADR-PAY-019/021/023/026; v1 no-extension; no direct gpi snapshot exposure
PIR-2024-07	PIR-2024-07 Wire Status Lite Pilot INC-2024-1182 Post-Implementation Review	Wire Status Lite failure, INC-2024-1182 impact, open lesson on status terminology

2.2 Agent configuration

- **Product skills** (Anthropic product-management plugin): brainstorm, competitive-brief, synthesize-research, write-spec (forked into a BRD template), roadmap-update, stakeholder-update.
- **estate-discovery skill** (custom): for every capability in a concept, resolve the data it needs -> producing system -> interface (existing or missing) -> owning team (latest effective date wins) -> governing policy clauses -> open Jira -> prior incidents.
- **Tools:** wiki_read / grep over the OpenWiki pages and graph_query over typed frontmatter (systems, endpoints, owners, policies).
- **Persona:** Sarah Whitfield, Senior Product Manager - Wire Tracking. Uploads PM-WTC-MR-0.3.

3. Graph model

The corpus yields **266 typed nodes** before counting document sections and individual fields. Counts below come from the canonical estate model used to generate the corpus.

Node type	Count	Examples
Person	49	Owners, approvers, engineers, compliance officers
Team / function	16	Engineering teams and 1st/2nd-line partners
System / module	13	CBO, SPS, CES, PPH, FCC, GPI-C, HMS, Sentinel, SanctionScreen, ENS, TDIP, CDP, IWB
API endpoint (existing)	20	v1/v2 REST, ENS, TDIP, CES, SPS, CBO BFF, step-up
API endpoint (missing / proposed)	5	Explicitly documented as not existing
Event topic / batch job	6	Kafka topics and the gpi batch pull
Jira issue / initiative	29	CBO, PPH, PNG, GTSI, FCT, ENS, TDA
Hold reason code	12	HRC-01 to HRC-12
Lifecycle state / status term	20	PPH v2 states + regulated status terms
gpi status / reason code	7	ACSP G000-G004, ACCC, RJCT
Dataset / feature table	9	TDIP datasets and feature tables
Model / EUA	4	M-FCT-0021, M-FCT-0034, M-TRS-0142, EUA-TRS-0007
Policy / standard	5	POL-FCC-014, DUS-07, MRM-POL-02, SEC-STD-22, FRS-02
Control	4	CTRL-PAY-012, -018, -031, -040
Regulation / guidance	10	SAR confidentiality, OFAC Part 501, UCC 4A, Reg J, FFIEC, FTC Act s5, GLBA, SR 26-2 ...
ADR	5	ADR-PAY-017 / 019 / 021 / 023 / 026

Node type	Count	Examples
Disclosure tier / copy key	11	P/C/G/R and approved copy
Notification event type / template	12	TRS. * catalog
Governance forum	6	ARB, Demand Board, FCT CAB, DRC, MRM, PIA
Incident / complaint	2	INC-2024-1182, CMP-2026-1189
Identifier type	6	cboRef, pphld, IMAD, OMAD, UETR, EndToEndId
Source document	15	Directory, memo, architecture, catalogs, designs, policies, Jira export, ADRs, PIR

3.1 Key relationship types

Edge	From -> To	Example
OWNS (valid_from, valid_to)	Team -> System	GTSI OWNS SYS-PNG-GPI valid_from 2026-10-01 (supersedes PNE, CNB-MEMO-2026-09)
EXPOSES / CONSUMES	System -> Endpoint/Topic	cbo-wire-bff CONSUMES EP-PPH-01 (v1)
HAS_FIELD / SOURCED_FROM	Endpoint -> Field -> System	EP-PPH-01.holdReasonDesc SOURCED_FROM HMS.description (Restricted)
COLLAPSES / MAPS_TO	Status -> State / Label	v1 PROCESSED COLLAPSES {RELEASED, SENT_TO_NETWORK, NETWORK_ACCEPTED, COMPLETED} MAPS_TO 'Completed'
GOVERNED_BY	Artifact -> Policy clause	Label 'Completed' GOVERNED_BY POL-FCC-014 s5.2
CLASSIFIED_AS / PERMITTED_PURPOSE	Dataset -> Class / Purpose	beneficiary_behavior_profile PERMITTED_PURPOSE 'FCT fraud only'
DEPENDS_ON / BLOCKS	Jira -> Jira	CBO-4471 DEPENDS_ON CBO-4473
SUNSETS_ON	Endpoint -> Date	EP-PPH-01 SUNSETS_ON 2027-03-31
CAUSED / RESULTED_IN	Incident -> ADR	INC-2024-1182 RESULTED_IN ADR-PAY-019
MISSING (gap node)	Capability -> Proposed endpoint	Client-safe hold status -> EP-HMS-X1 (FCT-1893)

4. The four headline cases

Each case requires joining facts from 4-7 documents written by different teams at different times. None of the source documents states the conclusion. That is the point of the demo.

Case 1 - The tooltip that would tip off (compliance violation + in-flight risk)

PM concept	FC-03: show the exact risk-system reason ('Unusual activity detected', 'Sanctions screening in progress'); Confirm/Reject on all held wires; ETA after confirmation.
Graph reveals	Tier R reasons (HRC-07..11, ~18.5% of holds) may never be shown or implied; R and G must look identical; no ETA for G/R; digital confirmation allowed only for HRC-01 (22% of holds) with step-up auth and <= \$5M; HRC-02 callback can never be confirmed in-channel. Worse: Jira CBO-4388 (merged, flag default ON, prod 2026-10-22) puts PPH v1 holdReasonDesc in a tooltip. That field is synced HMS free text - UAT comments show 'FRAUD_MODEL_HIGH sc=9xx' and 'SANCTIONS_REVIEW name match 0.91'.
Why search alone misses it	The Jira story never mentions policy; the policy never mentions the field; the API catalog only says the field is 'not curated'. Only the chain story -> field -> source system -> data classification -> policy clause exposes it.
Impact	Prevents a potential SAR-confidentiality breach (31 CFR 1020.320(e)) within two weeks; avoids building a non-compliant held-wire experience.

FC-03 —requires→ hold reason text
CBO-4388 —uses→ EP-PPH-01.holdReasonDesc —SOURCED_FROM→ HMS.description [Restricted] (JIRA-EXP-2026-10-05 → PPH-API-CAT-2026.3 → PPH-SYS-0VW-9.2 / PRSP-HMS-3.4)
HMS.description —contains→ HRC mnemonic, score band, list match (PRSP-HMS-3.4 s4)
HRC-07..11 —tier→ R —GOVERNED_BY→ POL-FCC-014 s5.1(1)(2), App. C (POL-FCC-014)
ADR-PAY-021 —forbids→ hold reasons leaving FCT boundary (ARB-PAY-REG-2026Q3)
FCT-2004 (open risk) —same field→ EP-PPH-01.holdReasonDesc (JIRA-EXP-2026-10-05)

Case 2 - 'Settled' is not a field (missing endpoint + semantics + hard deadline)

PM concept	FC-01: 'Settled' = CBO 'Completed', already available.
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Graph reveals	Wire Center consumes PPH v1, where PROCESSED collapses RELEASED, SENT_TO_NETWORK, NETWORK_ACCEPTED and COMPLETED; CBO-4402 maps it to 'Completed' - so 'Completed' appears at release, before the Fed accepts the wire. POL-FCC-014 s5.2 allows 'Completed/Settled' only after Fed acceptance (domestic) or gpi ACCC (international). The same mapping triggers ENS template TPL-WIRE-REL-02 'Wire completed'. Complaint CMP-2026-1189 is the consequence; PIR-2024-07 flagged this misunderstanding in 2024. Fixing it needs v2 (CBO-4471, unscheduled, v1 sunset 2027-03-31 with no extension), event-driven status (ADR-PAY-019), rail-aware mapping (Swift NETWORK_ACCEPTED is only a SwiftNet ACK) and PPH-2207 for the true Fed timestamp (no sponsor).
Why search alone misses it	Five documents each hold one piece: label mapping (CBO-ARCH-WC-4.1), state collapse (PPH-SYS-OVW-9.2 / PPH-API-CAT-2026.3), rail semantics (PPH-SYS-OVW-9.2), terminology rule (POL-FCC-014), deadline and missing field (PPH-API-CAT-2026.3 / JIRA-EXP-2026-10-05 / ARB-PAY-REG-2026Q3).
Impact	Turns a 'front-end' feature into a cross-team migration with a regulatory-complaint fix and a hard date on the critical path.

Case 3 - The FedEx map has no live feed and a new owner

PM concept	FC-02: live gpi route refreshed every minute; 'Expected delivery: Today'; engage Raj Malhotra (PNE).
Graph reveals	gpi data is pulled every 4 hours into an Oracle snapshot with no API; per-wire Tracker calls would exceed the 250k/month quota ~7x (PNG-1544 already rejected a smaller increase); ARB (2026-09-08) forbids exposing the snapshot directly; GTRS is not funded until 2027. The capability moved to GTSI (Elena Vasquez / Omar Siddiqui) on 2026-10-01 - the June directory, the gpi design doc, spike CBO-4480 and the PM's concept all still point to PNE. 8% of wires end at ACSP/G001 (non-gpi agent), where policy requires 'Tracking unavailable beyond [bank]' - so 'Expected delivery: Today' cannot be universal. TDA-2188 (hourly load) cannot make data fresher than its 4-hourly source. UETR is needed to join - only in v2 (ADR-PAY-023).
Why search alone misses it	Ownership requires temporal reasoning (memo dated after directory and design doc); freshness requires joining a batch schedule (PNG-TDD-6.0) to a downstream load (TDIP-CAT-2026.2).
Impact	Avoids a wasted month chasing the wrong team; resets the international scope to a realistic phase.

Case 4 - The insight that uses someone else's bank account

PM concept	FC-04: 'Beneficiary typically moves funds within 4 hours'; '92% of similar wires same day'; predicted delivery time; computed in CBO BFF.
Graph reveals	The only source for beneficiary post-receipt behavior is TDIP beneficiary_behavior_profile.bene_median_hours_to_outflow, derived from dda_txn_history (Restricted) of on-us beneficiaries, permitted purpose 'FCT fraud/mule detection only', input to Tier-1 model M-FCT-0034. Showing it to the originator breaches DUS-07 s4 (cross-client) and s3 (purpose limitation) and MRM-POL-02 s4. Cohort stats are allowed only above DUS-07 thresholds - EUR/DE passes, MXN/MX (largest client 21%) and NGN/NG fail. 'Predicted delivery' is a customer-facing estimate -> Tier 2 model -> 10-14 weeks validation + ~6-week queue. Domestic history needs Fed acceptance data that TDIP does not ingest (TDA-2210). ADR-PAY-026 forbids analytics in the BFF.
Why search alone misses it	The insight sentence contains no data-source name; the agent must infer the only producible source, then join its lineage, classification and purpose to two standards and the model inventory.
Impact	Removes a privacy violation from the design and replaces it with a compliant 'your own history' insight that can ship in Phase 1.

5. Run of show (about 15 minutes)

Time	Step	What to show
0:00	Context	The PM's concept looks complete. 'Mostly front-end, 1.5 Pls.'
1:00	Upload + Q1	Agent returns dependency map; flags outdated owner in the PM's own document
3:00	Q4 + Q5	Compliance review of FC-03 and the CBO-4388 alarm - the 'stop the release' moment
6:00	Q2 + Q3	'Settled' and 'live map' collapse into real constraints with dates
9:00	Q6	The beneficiary insight is caught; a compliant alternative is proposed
11:00	Q7 + Q10	Missing-endpoint table and the effort/engagement plan with hours by team
14:00	Close	Before: 'front-end, 1.5 Pls'. After: 16 teams/functions, ~1,900 hours, 3 phases, 4 avoided failures.

6. Ten demo questions with graph-grounded answers

Answers below are the target output. Section references point to source documents so presenters can click through. Paths show the traversal the agent performs.

Q1. Dependencies and owners

User

I've uploaded my market research and concept. Before I write the BRD: which existing systems does the Wire Tracking Center depend on, and who owns each today?

Agent: The concept depends on six systems outside the CBO portal, three CBO platform services, and eight partner functions that hold approval gates. One owner in your concept is out of date.

Capability	System	Owning team (current)	Contacts	Why
Wire lifecycle status	PRISM Payments Hub (SYS-PPH)	Payments Hub Engineering	Kevin O'Brien (EM), Sunita Rao (APIs), Laura Kim (PO)	v2 APIs + pay.wire.lifecycle.v2
Fed acceptance / settlement time	Fedwire Funds Connector (SYS-PNG-FFC)	Payment Networks Eng.	Brian Walsh, Chen Wei	net.fedwire.ack.v1 (OMAD)
International route	Swift Alliance & gpi Connector (SYS-PNG-GPI)	GTSI (since 2026-10-01)	Omar Siddiqui, Hannah Lindqvist, Tomasz Nowak	gpi snapshot / future GTRS
Hold status & client actions	PRSP Hold Management Service	Financial Crimes Technology	Grace Mensah, Daniel Kowalski	Client-safe status, attestation
Notifications	Enterprise Notification Service	Enterprise Notification Platform	Melissa Grant, Paul Henderson	New TRS.WIRE event types
Insights	Treasury Data & Insights Platform	Treasury Data & Analytics	Carlos Mendes, Dr. Aisha Rahman	Insights API (ADR-PAY-026)
Read model, entitlements, step-up	CBO SPS / CES / CBO Auth	CBO Platform & Entitlements	Nadia Haddad, Arjun Mehta	Reuse ACH Tracker pattern

- **Governance you must engage:** FCC Policy & Advisory (Jordan Ellis) for POL-FCC-014; Model Risk (Sophie Laurent); Privacy (Ethan Brooks); DRC (Patricia Moore); ARB (Nikhil Bose - next 2026-11-10, submit by 2026-10-27); InfoSec (Farah Ali); Payment Ops (Luis Ramirez, Tanya Brooks) and Treasury Support (Kim Nguyen) for readiness.
- **Correction:** your concept and spike CBO-4480 route gpi to Raj Malhotra (PNE). Memo CNB-MEMO-2026-09 moved gpi to GTSI effective 2026-10-01; PNE remains secondary on-call only until 2026-12-15 (GTSI-0112). The June directory (CNB-ORG-PTT-2026-06) and PNG-TDD-6.0 pre-date the change.
- **Not in your concept at all:** FCT, MRM, Privacy, FCC and ARB - each holds a gate on the critical path.

FC-02 -needs-> gpi status -PRODUCED_BY-> SYS-PNG-GPI
SYS-PNG-GPI -OWNED_BY{valid_to:2026-09-30}-> TEAM-PNE (CNB-ORG-PTT-2026-06 s4, PNG-TDD-6.0 s1)
SYS-PNG-GPI -OWNED_BY{valid_from:2026-10-01}-> TEAM-GTSI (CNB-MEMO-2026-09 s1) -> latest effective edge wins
CBO-4480 -MENTIONS-> Raj Malhotra -> flagged as stale reference

Sources: CNB-ORG-PTT-2026-06 s3-4; CNB-MEMO-2026-09 s1-4; CBO-ARCH-WC-4.1 s2; PNG-TDD-6.0 s1; JIRA-EXP-2026-10-05 CBO-4480.

Q2. Can we show 'Settled' for domestic wires?

User

My timeline ends in 'Settled'. Wire Center already shows 'Completed' - can we just use that?

Agent: No - and the current 'Completed' label is itself a problem.

1. **What 'Completed' means today:** Wire Center reads PPH v1, where PROCESSED covers RELEASED, SENT_TO_NETWORK, NETWORK_ACCEPTED and COMPLETED (PPH-SYS-OVW-9.2 s3). CBO-4402 (Feb 2026) labels PROCESSED as 'Completed', so clients see 'Completed' as soon as CNB releases the wire - before the Federal Reserve accepts it.
2. **Policy:** POL-FCC-014 s5.2 permits 'Completed/Settled' only after Fedwire acceptance (pacs.002 with OMAD) for domestic wires, or gpi ACCC for international. The existing label is non-compliant, and so is ENS template TPL-WIRE-REL-02 ('Wire completed: ...') triggered by the same status. Complaint CMP-2026-1189 / bug CBO-4419 is a real instance; PIR-2024-07 found 37% of users misread 'Processed' in 2024 and left an open action to validate terminology with FCC and Legal.

3. **What a true 'Settled' needs:** (a) PPH v2 + lifecycle events - CBO-4471 (34 pts, unscheduled) must finish before the **2027-03-31 v1 sunset (no extension, ARB 2026-09-08)**; (b) event-driven consumption via SPS (ADR-PAY-019 forbids polling; reuse CBO-3815) with a ~3-week topic ACL; (c) **rail-aware mapping** - Fedwire NETWORK_ACCEPTED = final settlement at the Fed; Swift NETWORK_ACCEPTED is only a SwiftNet ACK and must not be shown as settled; (d) the authoritative Fed timestamp needs **PPH-2207** (8 pts, backlog, no product sponsor) - otherwise use the stateHistory time, which is PPH processing time.
4. **Beneficiary credit cannot be shown for domestic wires:** the Fed does not report it (PNG-TDD-6.0 s2.2).

Recommended domestic milestones: Submitted -> Approved -> Released by CNB -> Sent to Federal Reserve -> **Delivered to beneficiary's bank** (Fed acceptance, show OMAD). Final copy via DRC. **Immediate action:** sponsor PPH-2207 at the Payments Platform Demand Board on 2026-10-21; raise the 'Completed' label and template with FCC as a live issue.

FC-01 'Settled' -equated_to-> CBO label 'Completed' -MAPS_FROM-> v1 PROCESSED (CBO-ARCH-WC-4.1 s3)
v1 PROCESSED -COLLAPSES-> {RELEASED, SENT_TO_NETWORK, NETWORK_ACCEPTED, COMPLETED} (PPH-SYS-OVW-9.2 s3)
'Completed' -GOVERNED_BY-> POL-FCC-014 s5.2 (only after Fed acceptance / gpi ACCC) => VIOLATION
Fed acceptance -CARRIED_ON-> net.fedwire.ack.v1 -CONSUMED_BY-> PPH -EXPOSED_IN-> v2 (approx.) | PPH-2207 (exact, MISSING)
EP-PPH-01 -SUNSETS_ON-> 2027-03-31 ; CBO-4471 status=unscheduled => deadline risk

Sources: CBO-ARCH-WC-4.1 s3, s5; PPH-SYS-OVW-9.2 s3-4; PPH-API-CAT-2026.3 s1, s3; PNG-TDD-6.0 s2.2; ENS-INT-3.2 s4; POL-FCC-014 s5.2; JIRA-EXP-2026-10-05 CBO-4402/4419/4471, PPH-2207; ARB-PAY-REG-2026Q3 minutes; PIR-2024-07 s3, s5.

Q3. Live international route like FedEx

User

For international wires I want the live SWIFT gpi route, refreshed every minute, with 'Expected delivery: Today'. Feasible for Q1 2027?

Agent: Not as specified. A 4-hour-fresh view is feasible in 2027-Q2; real-time depends on an unfunded 2027 initiative.

Constraint	Evidence
No gpi API exists	gpi updates are batch-pulled every 4 h into GPI_TRACKER_SNAPSHOT; consumers are Ops IWB and TDIP only (PNG-TDD-6.0 s3.2). EP 'GET /png/gpi/v1/...' does not exist.
Direct exposure not allowed	ARB 2026-09-08: exposing the snapshot table to channels is not approved; a read-only service with quota isolation needs ARB review.
Quota	Contract 250k Tracker calls/month at ~61% use; per-wire client lookups ~1.9M/month (~7x). PNG-1544 (4 h -> 2 h) already declined. Design must be change-feed + fan-out.
Real-time roadmap	GTRS (GTSI-0107) not funded for 2026; discovery 2027-Q2, build 2027-Q3/Q4.
Owner	GTSI since 2026-10-01 (not PNE); knowledge transfer until 2026-12-15.
Freshness trap	TDA-2188 moves TDIP's gpi load to hourly, but the source only refreshes every 4 h - data stays up to 4 h old.
Join key	UETR is required to join gpi data; only available in PPH v2 / events (ADR-PAY-023) - depends on CBO-4471.
Tracking gaps	8% of outbound wires end at ACSP/G001 (non-gpi agent). POL-FCC-014 s5.2 requires 'Tracking unavailable beyond [bank]' - no ETA.
Status words	'Delivered/Credited' only on gpi ACCC. Swift NETWORK_ACCEPTED is not delivery.

What you can promise: intermediary route, timestamps and deducted charges (reported for 34% of wires) for the client's own payments, labelled 'as of HH:MM - updated every 4 hours', served by a GTSI read-only service (~21 pts, ARB review).

Recommendation: move FC-02 to Phase 2 and give GTSI your requirements now so they shape GTRS discovery (Elena Vasquez asked for exactly this in GTSI-0107).

FC-02 -needs-> gpi status (UETR) -PRODUCED_BY-> BT-PNG-01 {every 4h} -WRITES-> GPI_TRACKER_SNAPSHOT
GPI_TRACKER_SNAPSHOT -READ_BY-> {IWB, TDIP nightly-hourly(TDA-2188)} => freshness bounded by 4h source
EP-PNG-X1 -status-> DOES NOT EXIST -PROPOSED_IN-> GTSI-0107 {2027}
ARB-2026-09-08 -PROHIBITS-> direct snapshot exposure
UETR -AVAILABLE_IN-> PPH v2 only -BLOCKED_BY-> CBO-4471

Sources: CNB-MEMO-2026-09; PNG-TDD-6.0 s3.2-3.5, Addendum A; TDIP-CAT-2026.2 s2, s5; POL-FCC-014 s5.2-5.3; JIRA-EXP-2026-10-05 GTSI-0107, TDA-2188; ARB-PAY-REG-2026Q3 ADR-PAY-023, minutes.

Q4. Review my held-wire design

User

Review FC-03. Clients want to know why a wire is held and to fix it themselves. Any issues?

Agent: Four policy issues and two missing interfaces. Here is a compliant redesign.

Design element	Finding	Rule
Show exact reason ('Unusual activity', 'Sanctions screening')	Prohibited for tier R (HRC-07..11 - fraud, ATO, sanctions, AML, legal: ~18.5% of holds). Words appear on the prohibited list.	POL-FCC-014 s5.1(1), App. C; 31 CFR 1020.320(e); OFAC Part 501 handling
Different treatment by reason	R and G must be indistinguishable: same label, icon, color, copy, actions and no timing.	POL-FCC-014 s5.1(2)
Confirm / Reject on all holds	Only HRC-01 duplicate-suspect (22%) allows digital attestation, with step-up auth; > \$5M still needs Wire Room release. HRC-02 callback (18%) can never be confirmed in-channel - offer 'Request callback'. HRC-03: admin approval; HRC-05: fund account; HRC-06: cancel and resubmit.	CTRL-PAY-031, CTRL-PAY-018; POL-FCC-014 App. A
'Released ~30 min after you confirm'	No ETA for G/R holds. For HRC-01 the observed median is 47 min - '30 min' would be inaccurate.	CTRL-PAY-040; POL-FCC-014 s5.1(3); FRS-02
Data source	Raw reasons cannot leave FCT; channels may not consume risk.hold.events.v1.	ADR-PAY-021; PRSP-HMS-3.4 s4

Missing interfaces (FCT): client-safe status GET /prsp/hms/v1/holds/{id}/client-view (FCT-1893, 21 pts, deprioritized because 'no channel consumer committed' - you are that consumer) and a client-attestation endpoint (none exists, ~13 pts). Plus CES entitlement WIRE_HOLD_ATTEST and reuse of step-up (EP-AUTH-01). **Effort:** 34 FCT pts x 8 h + 20% compliance testing = ~326 h; FCC review ~10 business days; FCT year-end freeze Dec-15 to Jan-05. **Business case:** pilot FCT-1951 resolved 82% of duplicate holds within 30 minutes.

Sources: PRSP-HMS-3.4 s3-7; POL-FCC-014 s4-5, App. A-C; JIRA-EXP-2026-10-05 FCT-1893/1951; ARB-PAY-REG-2026Q3 ADR-PAY-021; CNB-ORG-PTT-2026-06 s6.

Q5. Is anything in flight that conflicts?

User

Before I go further - is anything already in the Wire Center backlog or in production that conflicts with policy or with this feature?

Agent: Yes. One item needs action this week.

Priority	Item	Issue	Action / owner
Critical - before 2026-10-22	CBO-4388 hold tooltip (merged; flag WC_HOLD_TOOLTIP default ON; prod 2026-10-22)	Displays PPH v1 holdReasonDesc = HMS Restricted free text. UAT already shows fraud score bands and sanctions match text. Same field as open risk FCT-2004. AML_REVIEW descriptions could reveal FIU review (SAR confidentiality).	Turn flag off / pull story; notify Jordan Ellis (FCC) and Daniel Kowalski (FCT); Tom Becker (EM)
High	CBO-4402 'Completed' label (production since Feb 2026) + ENS TPL-WIRE-REL-02	'Completed' shown at release; template says 'Wire completed'. Conflicts with POL-FCC-014 s5.2; linked to CMP-2026-1189.	Interim relabel to 'Sent' and template change via DRC; FCC/Legal review of complaint; Marcus Chen
High	CBO-4471 v1 -> v2 migration	Unscheduled; PPH-2190 shows cbo-wire-bff 'Not started'; v1 ends 2027-03-31, no extension. Prerequisite for UETR, Settled, gpi join.	Commit in PI 27.1 (asks due 2026-11-06); present plan at ARB 2026-11-10
Medium	CBO-4480 gpi spike	Contact points to PNE; owner is now GTSI.	Re-route to Omar Siddiqui (Jira GTSI)
Low	TDA-2188 hourly gpi load	Will not improve freshness beyond 4 h source.	Set expectations in BRD

CBO-4388 {In Progress, release 2026-10-22} -USES→ EP-PPH-01.holdReasonDesc
 -SOURCED_FROM→ HMS.description [Restricted] -EXAMPLE→ 'FRAUD_MODEL_HIGH sc=9xx L1 queue' (PRSP-HMS-3.4 s4, JIRA-EXP-2026-10-05 comment)
 -GOVERNED_BY→ POL-FCC-014 s5.1 ; -CONFLICTS_WITH→ ADR-PAY-021 ; -SAME_AS→ FCT-2004 (open)
 ⇒ severity CRITICAL (time-bound: 14 days)

Sources: CBO-ARCH-WC-4.1 s3, s5; PPH-SYS-OVW-9.2 s6; PPH-API-CAT-2026.3 s1-2; PRSP-HMS-3.4 s4; ENS-INT-3.2 s4; POL-FCC-014; JIRA-EXP-2026-10-05 CBO-4388/4402/4471/4480, FCT-2004, PPH-2190; ARB-PAY-REG-2026Q3.

Q6. Can we show these insights and predictions?

User

FC-04: can we show 'Beneficiary typically moves funds within 4 hours', '92% of similar EUR wires complete same day', and a predicted delivery time based on the client's similar past wires?

Insight	Verdict	Why (graph evidence)	Compliant alternative
'Beneficiary typically moves funds within 4 hours'	Not permitted	Only source: TDIP beneficiary_behavior_profile.bene_median_hours_to_outflow, derived from dda_txn_history (Restricted) of on-us beneficiaries; permitted purpose FCT fraud/mule detection; input to Tier-1 model M-FCT-0034. Breaches DUS-07 s4 (cross-client) and s3 (purpose), MRM-POL-02 s4. For off-us beneficiaries no data exists.	'Your last N wires to this beneficiary were credited in a median of X hours' (client's own data, N >= 5, show basis)
'92% of similar EUR wires same day'	Conditionally permitted	DUS-07 s5 thresholds: >= 500 wires, >= 20 clients, largest client <= 15%. EUR/DE passes (31,200 / 1,140 / 4%). MXN/MX fails (21%), NGN/NG fails all. wire_corridor_stats_daily is Internal (Ops) and has no suppression flag. Descriptive = EUA (MRM-POL-02).	Serve via new TDIP corridor-stats endpoint with suppression; DRC wording + disclaimer
'Predicted delivery time'	Phase 3	Customer-facing estimate = Tier 2 model minimum: 10-14 weeks validation + ~6-week queue (Q4-2026). Earliest realistic production mid-2027.	Phase 1: descriptive 'typical' history (EUA). Phase 3: validated ETA

- **Architecture:** ADR-PAY-026 requires insights to be served by the TDIP Insights API - not computed in the CBO BFF as the concept assumes. Neither wire-history nor corridor endpoints exist; client_wire_history_features is an unregistered prototype.
- **Data gap:** domestic completion history needs Fed acceptance times, which TDIP does not ingest (TDA-2210, blocked on a PNE topic ACL). gpi history starts 2025-03-01.
- **Display rule:** never show insights on wires in a G or R hold (POL-FCC-014 s5.5(d)).
- **Effort:** TDIP 29 pts (~174 h: TDA-2210 8, wire-history endpoint 13, corridor endpoint 8); MRM EUA registration ~16 h; PIA ~4 weeks; DRC.

FC-04 'beneficiary moves funds' →requires→ beneficiary post-receipt outflow
 →ONLY_SOURCE→ beneficiary_behavior_profile.bene_median_hours_to_outflow (TDIP-CAT-2026.2 s3)
 →DERIVED_FROM→ dda_txn_history [Restricted] →PERMITTED_PURPOSE→ FCT fraud only
 →INPUT_TO→ M-FCT-0034 (Tier 1) →USE_LIMITED_BY→ MRM-POL-02 s4
 →GOVERNED_BY→ DUS-07 s3, s4 → VIOLATION (cross-client + purpose)

Sources: TDIP-CAT-2026.2 s2-5; DUS-07 s3-6; MRM-POL-02 s2-6; POL-FCC-014 s5.5; ARB-PAY-REG-2026Q3 ADR-PAY-026; JIRA-EXP-2026-10-05 TDA-2140/2188/2210.

Q7. Which endpoints are missing and who builds them?

User

List every endpoint, field or event we need that doesn't exist yet, the owning team, and any existing ticket.

Needed for	Missing interface / data	Owner	Ticket / status	Size
True Fed settlement time	PPH v2 settlement object (fedSettlementTs, settlementStatus by rail)	Payments Hub Eng.	PPH-2207 - backlog, no sponsor	8 pts
Event-driven wire status	SPS wire mapping module + ACL on pay.wire.lifecycle.v2	CBO Platform + PPH	New (pattern CBO-3815)	~13 pts + ~3 wk ACL
v2 adoption	cbo-wire-bff on v2; CES account-filter mapping	CBO WC + CBO Platform	CBO-4471 / CBO-4473	34 + 8 pts
Beneficiary search	v2 beneficiary-name index	Payments Hub Eng.	PPH-2251	13 pts
International route	Read-only gpi status service (interim) / GTRS (target)	GTSI	New / GTSI-0107 (2027)	~21 pts interim
Client-safe hold status	GET /prsp/hms/v1/holds/{id}/client-view	FCT	FCT-1893 - deprioritized	21 pts
Duplicate attestation	POST /prsp/hms/v1/holds/{id}/client-attestation	FCT	None	~13 pts
Own-history insight	GET /tdip/insights/v1/wire-history/{clientId}	TDIP	None	13 pts
Corridor insight	GET /tdip/insights/v1/corridor-stats (DUS-07 suppression)	TDIP	None (TDA-2140 basis)	8 pts
Domestic history	fed_ack_events dataset + topic ACL	TDIP + PNE	TDA-2210 - backlog	8 pts
Notifications	Event types: DELIVERED_TO_BENE_BANK, CREDITED (gpi), RETURNED, HOLD_ACTION_REQUIRED, GPI_MILESTONE	ENS + CBO	None	5 x ~12 ENS h

Needed for	Missing interface / data	Owner	Ticket / status	Size
Entitlements	WIRE_HOLD_ATTEST; WIRE_STATUS_SHARE (Phase 3)	CBO Platform + InfoSec	None	~5 pts

Sources: CBO-ARCH-WC-4.1 s7-8; PPH-API-CAT-2026.3 s3, s6; PNG-TDD-6.0 s3.2; PRSP-HMS-3.4 s5; ENS-INT-3.2 s3-4; TDIP-CAT-2026.2 s4-5; JIRA-EXP-2026-10-05.

Q8. 'Notify me when this wire settles'

User

Can a client subscribe to a single wire - 'notify me when it settles'?

- **Not natively:** ENS subscriptions are by event type and account; per-entity subscriptions arrive with ENS-1120 in 2027-H1.
- **No wire milestone events exist** beyond APPROVAL_REQUIRED, RELEASED and REJECTED (and TPL-WIRE-REL-02 wording must change - see Q2).
- **Workaround that works now:** ENS 'producer-resolved recipients' - CBO SPS keeps a per-user watch list and sends a registered event to named users.
- **Lead time:** ENS onboarding SLA 6 weeks incl. DRC; submissions after 2026-11-20 slip past the year-end freeze.
- **Policy:** a 'notify me when released' option on held wires is acceptable only if offered identically for every hold tier, with no timing (POL-FCC-014 s5.1(2-3)).
- **Effort:** ENS ~60 h (5 event types); CBO watch list + producer ~13 pts (~85 h).

Sources: ENS-INT-3.2 s2-4, s6; POL-FCC-014 s5.1; JIRA-EXP-2026-10-05 ENS-1120.

Q9. Has this been tried before? What can we reuse?

User

Has anyone attempted wire tracking before? What should we avoid, and what can we reuse?

Prior work	What happened	Implication
Wire Status Lite (CBO-3120, 2024)	30-second polling of PPH v1 caused INC-2024-1182: 1,240 wires delayed 47 min, 312 missed the Fed cutoff, \$41,800 compensation to 14 clients. Epic closed.	Hard rule ADR-PAY-019: no polling; v1 limited to 20 TPS. Design must be event-driven.
PIR-2024-07 open action	'Validate status terminology with FCC and Legal' never closed; CBO-4402 later introduced 'Completed'.	Make FCC/Legal terminology review an explicit BRD gate.
ACH Payment Tracker (CBO-3790, 2025)	Delivered <cbo-journey-timeline> (CBO-3802) and SPS (CBO-3815) with ENS events.	Reuse both: ~30 pts (~195 h) saved vs building new; SPS is payment-type agnostic by design.
Duplicate attestation pilot (FCT-1951, 2026)	82% of duplicate holds resolved within 30 min; CTRL-PAY-031 now allows digital attestation.	Strong evidence for the HRC-01 self-service flow.
Treasury Insights cash forecast (2025)	Insights API pattern (EP-TDIP-01); model validated in 9 weeks as Tier 3.	Reuse the serving pattern; note re-tiering to Tier 2 under MRM v7.0 - plan 10-14 weeks for any ETA model.

Sources: PIR-2024-07; JIRA-EXP-2026-10-05 CBO-3120/3790/3802/3815, FCT-1951; ARB-PAY-REG-2026Q3 ADR-PAY-019; TDIP-CAT-2026.2 s4; MRM-POL-02 s5.

Q10. Effort, engagement plan and critical path for the BRD

User

Draft the dependency, effort and engagement section of the BRD: which teams to talk to, how many hours to allocate, approvals, lead times and the critical path.

Agent: Estimates apply the planning factors in CNB-ORG-PTT-2026-06 s6 to the story points in Jira and the sizes above. Your concept assumed mostly CBO front-end work; the graph shows 16 teams/functions and ~1,900 hours for Phases 1-2.

Team / function	Scope	Size	Hours	Lead time / gate	Contact
CBO Wire Center squad	CBO-4471 (34), tracker UI with timeline reuse (21), held-wire UX (13), insights UI (8), notification prefs / watch list (13), label + copy fixes (3)	92 pts	598	PI 27.1 planning 2026-11-17/18	Tom Becker

Team / function	Scope	Size	Hours	Lead time / gate	Contact
CBO Platform & Entitlements	CBO-4473 (8), SPS wire mapping (13), CES entitlements (5)	26 pts	169	Topic ACL ~3 wks	Nadia Haddad
Payments Hub Eng.	PPH-2207 (8), PPH-2251 (13), ACL / v2 consumer support (~24 h)	21 pts	192	Demand Board 2026-10-21; 90% committed	Kevin O'Brien
Payment Networks Eng.	net.fedwire.ack.v1 ACL for TDIP; consult	-	16	Change windows	Brian Walsh
GTSI	Read-only gpi status service (21) + ARB prep	21 pts	163	KT until 2026-12-15; ARB	Omar Siddiqui
Financial Crimes Tech.	FCT-1893 (21) + attestation endpoint (13) + 20% compliance testing	34 pts	326	FCT CAB; freeze Dec-15 to Jan-05	Grace Mensah
Enterprise Notification	5 new TRS.WIRE event types	-	60	6-week SLA; submit before 2026-11-20	Melissa Grant
TDIP	TDA-2210 (8), wire-history endpoint (13), corridor endpoint (8)	29 pts	174	PI 27.1	Carlos Mendes
Model Risk Mgmt	EUA registration (Phase 1). Tier 2 ETA validation (Phase 3): 160-240 h	-	16	~2 wks (EUA); 16-20 wks (Tier 2)	Sophie Laurent
FCC Policy & Advisory	Two reviews: held-wire design, status terminology	-	40	~10 business days each	Jordan Ellis
Legal / DRC	Copy, disclaimers, templates	-	16	Bi-weekly Thursdays	Patricia Moore
Privacy Office	PIA for insights	-	24	~4 weeks	Ethan Brooks
Information Security	Step-up on attestation; entitlement review	-	32	3-week SLA	Farah Ali
Enterprise Architecture	ARB submission (v1 migration plan + design)	-	12	ARB 2026-11-10 (submit by 2026-10-27)	Nikhil Bose
Payment Operations	SOPs: attestation queue, >\$5M release, repair	-	40	Ops Readiness 4 wks pre go-live	Luis Ramirez / Tanya Brooks
Commercial Service Center	Scripts and training	-	24	3 wks pre go-live	Kim Nguyen
Total (Phases 1-2)			~1,900	+160-240 h MRM in Phase 3	

Critical path

1. **PPH v1 sunset 2027-03-31** - CBO-4471 + CBO-4473 must land in PI 27.1 (ends 2027-03-05). Everything else (UETR, Settled, gpi join) sits on it.
2. **Topic ACL + SPS wire mapping** (~3 weeks lead) - prerequisite for any live status.
3. **FCT capacity and freeze** - FCT-1893 and attestation must clear FCC sign-off; no deploys Dec-15 to Jan-05.
4. **GTSI knowledge transfer** to 2026-12-15 - interim gpi service realistically PI 27.2.
5. **MRM queue + Tier 2 validation** (~16-20 weeks) - predictive ETA cannot be in the MVP.

Recommended phasing

Phase	Window	Scope
1 - MVP	PI 27.1 (launch ~2027-03)	v2 migration; domestic milestones to 'Delivered to beneficiary's bank'; corrected labels and templates; tier-compliant held-wire view; HRC-01 attestation (if FCT capacity); returns with ISO reason; category notifications; own-history 'typical time' insight (EUA)
2	PI 27.2	International view from GTSI read-only service ('as of', 4-hour freshness, G001 handling, deducted charges); corridor stats with suppression; per-wire watch notifications
3	2027-H2	Validated predictive ETA (Tier 2); real-time gpi via GTRS; ENS-1120 entity subscriptions; share-with-recipient link after SEC-STD-22 review and PIA

Actions for this week

- Stop CBO-4388 before the 2026-10-22 release; inform FCC (Jordan Ellis) and FCT (Daniel Kowalski).
- Raise the 'Completed' label and TPL-WIRE-REL-02 with FCC and DRC; link to CMP-2026-1189.
- Sponsor PPH-2207 at the Demand Board (2026-10-21); request topic ACLs.
- Re-route CBO-4480 to GTSI; share requirements for GTSI-0107 discovery.
- Book ARB 2026-11-10 (submit by 2026-10-27); submit dependency asks for PI 27.1 by 2026-11-06; file ENS onboarding before 2026-11-20.

Appendix A - Seeded findings (answer key for scoring the agent)

Use this list to score recall. A strong graph-grounded agent should surface at least 12 of 16 across the 10 questions, with correct sources. Hops = number of documents that must be joined.

ID	Finding	Docs to join	Hops	Severity
F1	CBO-4388 will expose Restricted HMS text via v1 holdReasonDesc (release 2026-10-22)	JIRA-EXP-2026-10-05; PPH-API-CAT-2026.3; PPH-SYS-OVW-9.2; PRSP-HMS-3.4; POL-FCC-014; ARB-PAY-REG-2026Q3	6	Critical
F2	'Completed' shown at release; TPL-WIRE-REL-02 'Wire completed'; complaint CMP-2026-1189	CBO-ARCH-WC-4.1; PPH-SYS-OVW-9.2; PPH-API-CAT-2026.3; ENS-INT-3.2; POL-FCC-014; JIRA-EXP-2026-10-05; PIR-2024-07	7	High
F3	True 'Settled' needs v2, rail-aware mapping, PPH-2207 (no sponsor)	CBO-ARCH-WC-4.1; PPH-SYS-OVW-9.2; PPH-API-CAT-2026.3; PNG-TDD-6.0; POL-FCC-014; JIRA-EXP-2026-10-05	6	High
F4	v1 sunset 2027-03-31 (no extension) vs unscheduled CBO-4471	PPH-API-CAT-2026.3; JIRA-EXP-2026-10-05; ARB-PAY-REG-2026Q3; CNB-ORG-PTT-2026-06	4	High
F5	No gpi API; 4-hourly batch; quota ~7x; ARB forbids direct exposure; GTRS 2027	PNG-TDD-6.0; ARB-PAY-REG-2026Q3; JIRA-EXP-2026-10-05; CNB-MEMO-2026-09	4	High
F6	gpi ownership moved PNE -> GTSI (2026-10-01); concept and CBO-4480 stale	CNB-ORG-PTT-2026-06; CNB-MEMO-2026-09; PNG-TDD-6.0; JIRA-EXP-2026-10-05	4	Medium
F7	8% of wires end at ACSP/G001; 'Expected delivery: Today' cannot be universal	PNG-TDD-6.0; POL-FCC-014	2	Medium
F8	TDA-2188 hourly load cannot beat the 4-hour source	TDIP-CAT-2026.2; PNG-TDD-6.0	2	Low
F9	Exact hold reasons + Confirm/Reject for all holds violate policy/controls	PRSP-HMS-3.4; POL-FCC-014; ARB-PAY-REG-2026Q3; JIRA-EXP-2026-10-05	4	Critical
F10	Beneficiary outflow insight = cross-client + purpose limitation + Tier-1 model use	TDIP-CAT-2026.2; DUS-07; MRM-POL-02	3	Critical
F11	Corridor stats need DUS-07 suppression (MXN, NGN fail)	TDIP-CAT-2026.2; DUS-07	2	Medium
F12	Predicted delivery = Tier 2 model; 16-20 weeks incl. queue	MRM-POL-02; TDIP-CAT-2026.2; ARB-PAY-REG-2026Q3	3	High (timeline)
F13	Domestic insight history blocked by TDA-2210 (Fed acks not ingested)	TDIP-CAT-2026.2; PNG-TDD-6.0; JIRA-EXP-2026-10-05	3	Medium
F14	Per-wire notifications unsupported until ENS-1120; 6-week onboarding; freeze	ENS-INT-3.2; JIRA-EXP-2026-10-05; CNB-ORG-PTT-2026-06	3	Medium
F15	Share-with-recipient link must exclude accounts/fees, expire <= 7 days, needs SEC-STD-22 + PIA	POL-FCC-014; CBO-ARCH-WC-4.1	2	Medium
F16	Planning constraints: PPH 90% committed, FCT freeze, GTSI KT, PI dates	CNB-ORG-PTT-2026-06; CNB-MEMO-2026-09; JIRA-EXP-2026-10-05	3	Planning

Appendix B - Making the graph work on this corpus

- **Entity IDs:** extract with patterns such as SYS-*, EP-*, EV-*, BT-*, HRC-\d\d, POL-*, DUS-*, MRM-*, ADR-PAY-\d+, CTRL-PAY-\d+, Jira keys [A-Z]+\d+, INC-*, CMP-*, M-FCT-*, M-TRS-*.
- **Typed frontmatter** per wiki page (type, id, owned_by, exposes, consumes, sourced_from, classification, permitted_purpose, governed_by, status, sunset_date, valid_from) so traversal does not depend on prose.
- **Temporal precedence:** when two OWNS edges conflict, prefer the latest valid_from (memo 2026-09 over directory 2026-06 and design 2025-11).
- **Gap nodes:** represent 'DOES NOT EXIST' interfaces as nodes linked to the capability and to their proposal ticket - this is what lets the agent answer 'what is missing'.
- **Field-level lineage:** model holdReasonDesc and bene_median_hours_to_outflow as nodes; Cases 1 and 4 depend on field-level edges.
- **estate-discovery prompt:** 'For each capability: data needed -> producing system -> interface (exists?) -> current owner -> governing clauses -> open tickets -> prior incidents. Flag conflicts, stale references and deadlines.'

Before / after (closing slide)

Before: 'Mostly front-end work for the CBO Wire Center squad; 1.5 Pls; GA Q1 2027 with FC-01 to FC-07.'

After: 16 teams/functions, ~1,900 hours for Phases 1-2, a hard 2027-03-31 deadline on the critical path, one production compliance risk stopped two weeks before release, two design violations removed, and a three-phase plan with named owners and dates.